

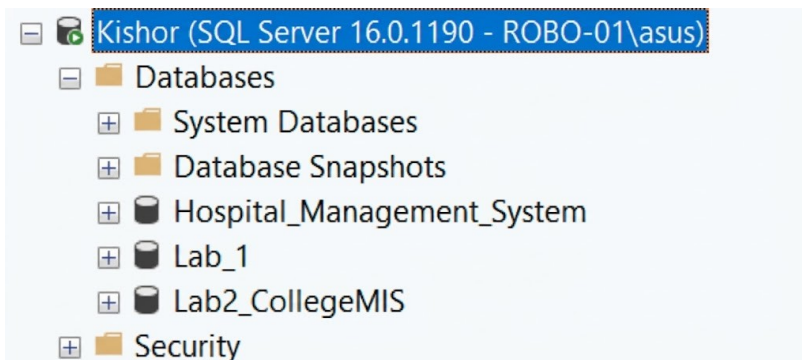
Database: Hospital_Management_System

1. Create database.

Query:

```
CREATE DATABASE Hospital_Management_System;
```

Output:

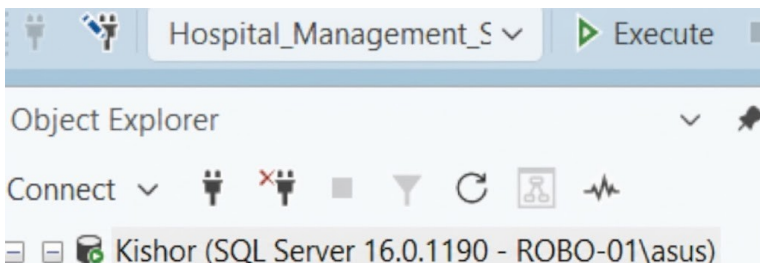


2. Use database.

Query:

```
USE Hospital_Management_System;
```

Output:



3. Create above 4 tables with given constraints.

Query:

```
CREATE TABLE Department(  
did INT IDENTITY(1,1) PRIMARY KEY,  
dname VARCHAR(20),  
location VARCHAR(30)  
);
```

```
CREATE TABLE Doctor(  

```

```
docid INT IDENTITY(1,1) PRIMARY KEY,  
dname VARCHAR(20) NOT NULL,  
gender CHAR(1)  
CHECK (gender IN('M','F')),  
specialization VARCHAR(20),  
phone VARCHAR(10),  
salary DECIMAL(10,2)  
CHECK (salary>50000.00),  
did INT,  
FOREIGN KEY(did) REFERENCES Department(did)  
);
```

```
CREATE TABLE Patient(  
pid INT IDENTITY(1,1) PRIMARY KEY,  
pname VARCHAR(20),  
age INT,  
CHECK (age >= 0),  
gender CHAR(1),  
CHECK (gender IN('M','F')),  
phone VARCHAR(10) UNIQUE,  
Rdate DATE DEFAULT GETDATE()  
);
```

```
CREATE TABLE Appointment(  
aid INT IDENTITY(1,1) PRIMARY KEY,  
pid INT,  
FOREIGN KEY(pid) REFERENCES Patient(pid),  
docid INT,  
FOREIGN KEY(docid) REFERENCES Doctor(docid),  
appdate DATE DEFAULT GETDATE(),  
status VARCHAR(20),  
CHECK (status IN('pending','completed','cancelled'))  
);
```

Output:

Kishor (SQL Server 16.0.1190 - ROBO-01\asus)

Databases

System Databases

Database Snapshots

Hospital_Management_System

Database Diagrams

Tables

System Tables

FileTables

External Tables

Graph Tables

dbo.Department

Columns

did (PK, int, not null)

dname (varchar(20), null)

location (varchar(30), null)

dbo.Doctor

Columns

docid (PK, int, not null)

dname (varchar(20), not null)

gender (char(1), null)

specialization (varchar(20), null)

phone (varchar(10), null)

salary (decimal(10,2), null)

did (FK, int, null)

dbo.Patient

Columns

pid (PK, int, not null)

pname (varchar(20), null)

age (int, null)

gender (char(1), null)

phone (varchar(10), null)

Rdate (date, null)

dbo.Appointment

Columns

aid (PK, int, not null)

pid (FK, int, null)

docid (FK, int, null)

appdate (date, null)

status (varchar(20), null)

4. Insert any 10/10 valid records to each of the table.

Query:

```
INSERT INTO Department (dname, location)
VALUES
('Cardiology', 'OPD Block, 2nd Floor'),
('Pediatrics', 'Kanti Children Block'),
('Neurology', 'New Emergency Building'),
('Orthopedics', 'Trauma Centre, Ground Floor'),
('Dermatology', 'Old Block, Room 104'),
('Oncology', 'Cancer Wing, 4th Floor'),
('Gastroenterology', 'OPD Block, 1st Floor'),
('Psychiatry', 'Mental Health Unit, Block E'),
('Ophthalmology', 'Eye Clinic, Ground Floor'),
('General Medicine', 'Main Building, 1st Floor');
SELECT *
FROM Department;
```

Output:

Results		Messages	
	did	dname	location
1	1	Cardiology	OPD Block, 2nd Floor
2	2	Pediatrics	Kanti Children Block
3	3	Neurology	New Emergency Building
4	4	Orthopedics	Trauma Centre, Ground Floor
5	5	Dermatology	Old Block, Room 104
6	6	Oncology	Cancer Wing, 4th Floor
7	7	Gastroenterology	OPD Block, 1st Floor
8	8	Psychiatry	Mental Health Unit, Block E
9	9	Ophthalmology	Eye Clinic, Ground Floor
10	10	General Medicine	Main Building, 1st Floor

```
INSERT INTO Doctor (dname, gender, specialization, phone, salary, did)
VALUES
('Dr. Ram Shrestha', 'M', 'Cardiologist', '9851012345', 125000.00, 1),
('Dr. Sita Thapa', 'F', 'Pediatrician', '9841123456', 95000.00, 2),
('Dr. Rajesh Adhikari', 'M', 'Neurologist', '9803123457', 145000.00, 3),
('Dr. Deepa Rai', 'F', 'Orthopedic Surgeon', '9818123458', 115000.00, 4),
('Dr. Sunil Joshi', 'M', 'Dermatologist', '9860123459', 88000.00, 5),
('Dr. Anju Sharma', 'F', 'Oncologist', '9849123460', 150000.00, 6),
('Dr. Binod Pradhan', 'M', 'Gastroenterologist', '9801123461', 105000.00, 7),
('Dr. Pooja Karki', 'F', 'Psychiatrist', '9851234562', 92000.00, 8),
('Dr. Hari Lama', 'M', 'Ophthalmologist', '9842123463', 89000.00, 9),
```

```
('Dr. Nabina Maharjan', 'F', 'General Physician', '9813123464', 78000.00, 10);
SELECT *
FROM Doctor;
```

Output:

Results		Messages					
	docid	dname	gender	specialization	phone	salary	did
1	1	Dr. Ram Shrestha	M	Cardiologist	9851012345	125000.00	1
2	2	Dr. Sita Thapa	F	Pediatrician	9841123456	95000.00	2
3	3	Dr. Rajesh Adhikari	M	Neurologist	9803123457	145000.00	3
4	4	Dr. Deepa Rai	F	Orthopedic Surgeon	9818123458	115000.00	4
5	5	Dr. Sunil Joshi	M	Dermatologist	9860123459	88000.00	5
6	6	Dr. Anju Sharma	F	Oncologist	9849123460	150000.00	6
7	7	Dr. Binod Pradhan	M	Gastroenterologist	9801123461	105000.00	7
8	8	Dr. Pooja Karki	F	Psychiatrist	9851234562	92000.00	8
9	9	Dr. Hari Lama	M	Ophthalmologist	9842123463	89000.00	9
10	10	Dr. Nabina Maharjan	F	General Physician	9813123464	78000.00	10

```
INSERT INTO Patient (pname, age, gender, phone, Rdate)
VALUES
('Santosh Tamang', 28, 'M', '9841987654', '2026-06-11'),
('Kriti Rajbhandari', 45, 'F', '9801987653', DEFAULT),
('Aayush Bhattarai', 12, 'M', '9860987652', '2026-06-19'),
('Saraswoti Pokharel', 34, 'F', '9818987651', '2026-06-15'),
('Prem Bahadur Basnet', 60, 'M', '9851087650', DEFAULT),
('Manisha Ghimire', 22, 'F', '9849087649', '2026-06-21'),
('Ganesh Gurung', 71, 'M', '9803087648', DEFAULT),
('Anjali Shahi', 5, 'F', '9813087647', '2026-06-04'),
('Ramesh Khadka', 52, 'M', '9861087646', DEFAULT),
('Sunita Chaudhary', 41, 'F', '9842087645', DEFAULT);
SELECT *
FROM Patient;
```

Output:

Results		Messages				
	pid	pname	age	gender	phone	Rdate
1	1	Santosh Tamang	28	M	9841987654	2026-06-11
2	2	Kriti Rajbhandari	45	F	9801987653	2026-06-30
3	3	Aayush Bhattarai	12	M	9860987652	2026-06-19
4	4	Saraswoti Pokharel	34	F	9818987651	2026-06-15
5	5	Prem Bahadur Basnet	60	M	9851087650	2026-06-30
6	6	Manisha Ghimire	22	F	9849087649	2026-06-21
7	7	Ganesh Gurung	71	M	9803087648	2026-06-30
8	8	Anjali Shahi	5	F	9813087647	2026-06-04
9	9	Ramesh Khadka	52	M	9861087646	2026-06-30
10	10	Sunita Chaudhary	41	F	9842087645	2026-06-30

```

INSERT INTO Appointment (pid, docid, appdate, status)
VALUES
(1, 1, '2026-06-23', 'completed'),
(2, 3, DEFAULT, 'pending'),
(3, 2, '2026-06-21', 'completed'),
(4, 4, DEFAULT, 'cancelled'),
(5, 6, DEFAULT, 'pending'),
(6, 5, DEFAULT, 'completed'),
(7, 7, DEFAULT, 'completed'),
(8, 2, DEFAULT, 'pending'),
(9, 10, DEFAULT, 'pending'),
(10, 8, DEFAULT, 'cancelled');
SELECT *
FROM Appointment;

```

Output:

	aid	pid	docid	appdate	status
1	1	1	1	2026-06-23	completed
2	2	2	3	2026-06-30	pending
3	3	3	2	2026-06-21	completed
4	4	4	4	2026-06-30	cancelled
5	5	5	6	2026-06-30	pending
6	6	6	5	2026-06-30	completed
7	7	7	7	2026-06-30	completed
8	8	8	2	2026-06-30	pending
9	9	9	10	2026-06-30	pending
10	10	10	8	2026-06-30	cancelled

5. Try to insert patient with age -20.

Query:

```

INSERT INTO Patient (pname, age, gender, phone, Rdate)
VALUES
('Ram', -20, 'M', '9841893528', '2026-06-24');

```

Output:

Messages
Msg 547, Level 16, State 0, Line 1 The INSERT statement conflicted with the CHECK constraint "CK_Patient_age_5165187F". The conflict occurred in database "Hospital_Management_System". The statement has been terminated. Completion time: 2026-06-30T21:51:01.5362649+05:45

6. Display all doctors with their department names.

Query:

```
SELECT d.dname, dept.dname as department_name
FROM Doctor d
INNER JOIN
Department dept
ON d.did = dept.did;
```

Output:

	dname	department_name
1	Dr. Ram Shrestha	Cardiology
2	Dr. Sita Thapa	Pediatrics
3	Dr. Rajesh Adhikari	Neurology
4	Dr. Deepa Rai	Orthopedics
5	Dr. Sunil Joshi	Dermatology
6	Dr. Anju Sharma	Oncology
7	Dr. Binod Pradhan	Gastroenterology
8	Dr. Pooja Karki	Psychiatry
9	Dr. Hari Lama	Ophthalmology
10	Dr. Nabina Maharjan	General Medicine

7. Find all patients registered today.

Query:

```
SELECT pid,pname,age,gender,phone,Rdate
FROM Patient
WHERE Rdate = CAST(GETDATE() AS DATE);
```

Output:

	pid	pname	age	gender	phone	Rdate
1	2	Kriti Rajbhandari	45	F	9801987653	2026-06-30
2	5	Prem Bahadur Basnet	60	M	9851087650	2026-06-30
3	7	Ganesh Gurung	71	M	9803087648	2026-06-30
4	9	Ramesh Khadka	52	M	9861087646	2026-06-30
5	10	Sunita Chaudhary	41	F	9842087645	2026-06-30

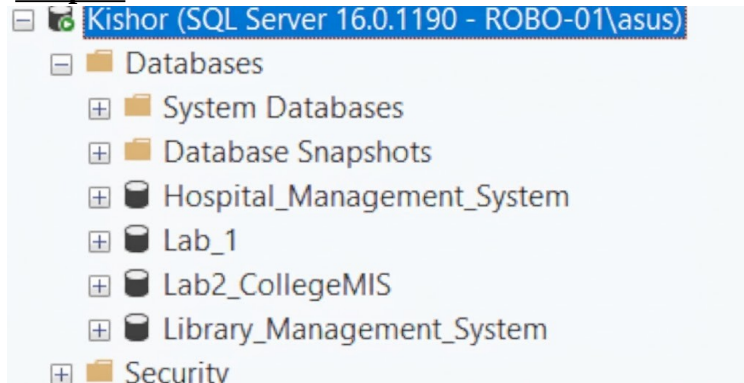
Database: Library_Management_System

1. Create database

Query:

```
CREATE DATABASE Library_Management_System;
```

Output:

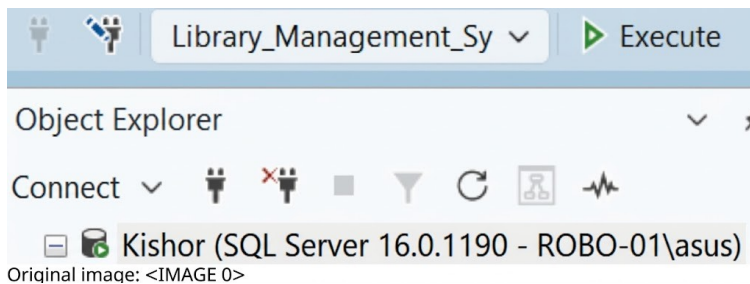


2. Use database.

Query:

```
USE Library_Management_System;
```

Output:



Original image: <IMAGE 0>

3. Create above 3 tables with suitable data types and required constraints.

Query:

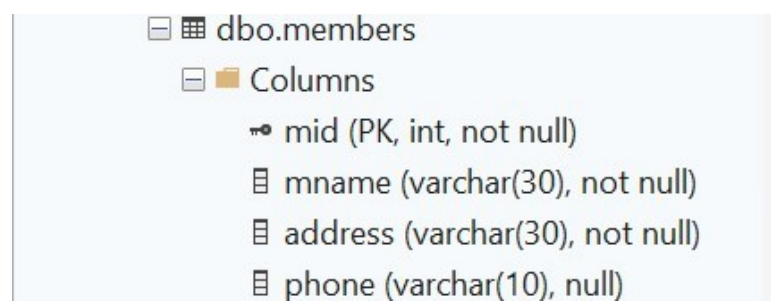
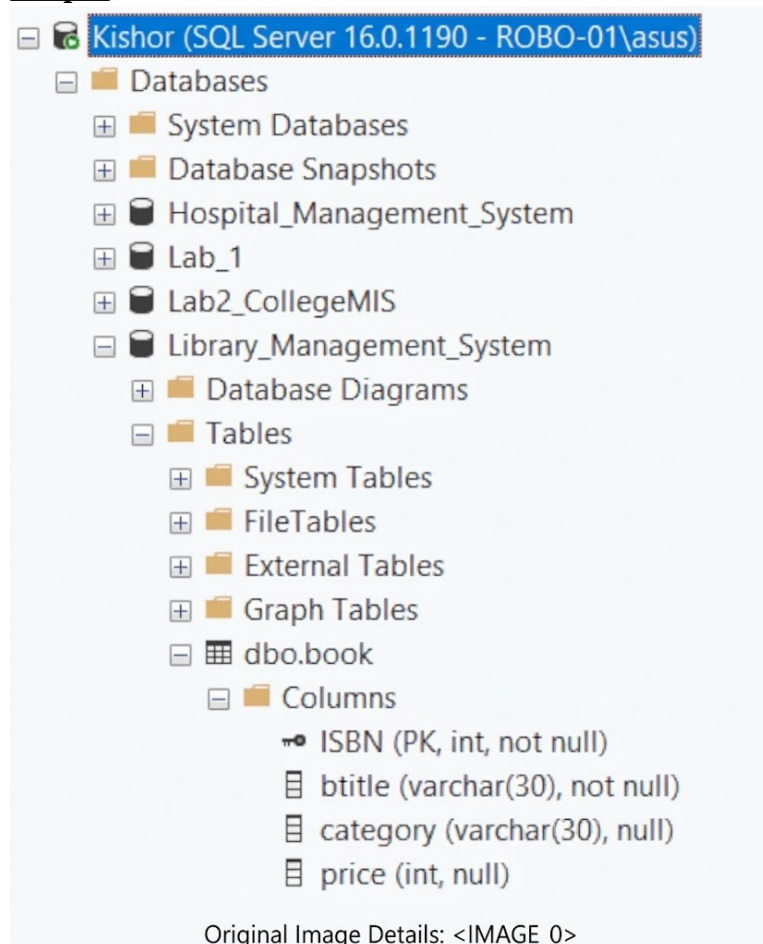
```
CREATE TABLE book(  
ISBN INT IDENTITY (100,1) PRIMARY KEY,  
btitle VARCHAR(30) NOT NULL,  
category VARCHAR(30),  
price INT,  
nop INT  
);
```

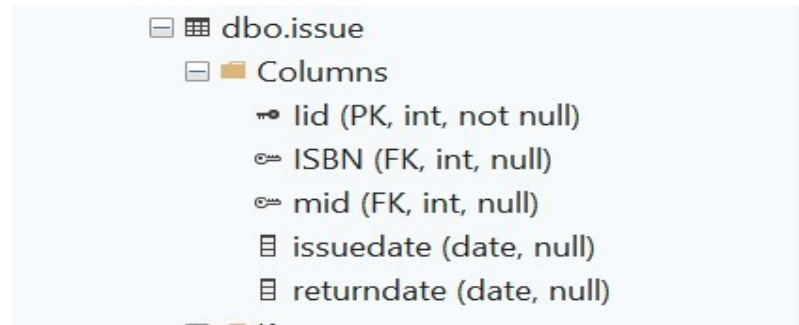
```

CREATE TABLE members(
mid INT IDENTITY(1000,1) PRIMARY KEY,
mname VARCHAR(30) NOT NULL,
address VARCHAR(30) NOT NULL,
phone VARCHAR(10) UNIQUE
);
CREATE TABLE issue(
lid INT IDENTITY(5000,1) PRIMARY KEY,
ISBN INT,
FOREIGN KEY (ISBN) REFERENCES book(ISBN),
mid INT,
FOREIGN KEY (mid) REFERENCES members(mid),
issuedate DATE DEFAULT GETDATE(),
returndate DATE NULL
);

```

Output





4. Insert any 10 records to each of the tables.

Query:

```
INSERT INTO book (btitle, category, price, nop)
VALUES
('Introduction to Algorithms', 'Computer Science', 850, 600),
('Learning Web Development', 'Web Design', 320, 250),
('Cloud Computing Bible', 'Infrastructure', 750, 480),
('Network Security Essentials', 'Security', 580, 310),
('Discrete Mathematics', 'Mathematics', 420, 290),
('The C++ Programming Language', 'Programming', 690, 540),
('Deep Learning Textbook', 'AI & Data Science', 980, 620),
('Compiler Design Foundations', 'Computer Science', 510, 330),
('The Hobbit', 'Fantasy Fiction', 350, 310),
('The Great Gatsby', 'Classic Literature', 280, 180);
SELECT *
FROM book;
```

Output:

Results		Messages			
	ISBN	btitle	category	price	nop
1	100	Introduction to Algorithms	Computer Science	850	600
2	101	Learning Web Development	Web Design	320	250
3	102	Cloud Computing Bible	Infrastructure	750	480
4	103	Network Security Essentials	Security	580	310
5	104	Discrete Mathematics	Mathematics	420	290
6	105	The C++ Programming Language	Programming	690	540
7	106	Deep Learning Textbook	AI & Data Science	980	620
8	107	Compiler Design Foundations	Computer Science	510	330
9	108	The Hobbit	Fantasy Fiction	350	310
10	109	The Great Gatsby	Classic Literature	280	180

```

INSERT INTO members (mname, address, phone)
VALUES
('Rajesh Khadka', 'Lalitpur', '9812345678'),
('Pooja Shrestha', 'Bhaktapur', '9841234567'),
('Sandeep Tamang', 'Kathmandu', '9803456789'),
('Aayusha Adhikari', 'Kirtipur', '9851987654'),
('Niranjan Thapa', 'Pokhara', '9865432109'),
('Deepa Magar', 'Dharan', '9741122334'),
('Suresh Chaudhary', 'Biratnagar', '9802233445'),
('Manisha Ghimire', 'Chitwan', '9813344556'),
('Bikram Shah', 'Butwal', '9844455667'),
('Kritika Pandey', 'Hetauda', '9867788990');
SELECT *
FROM members;

```

Output:

Results		Messages		
	mid	mname	address	phone
1	1000	Rajesh Khadka	Lalitpur	9812345678
2	1001	Pooja Shrestha	Bhaktapur	9841234567
3	1002	Sandeep Tamang	Kathmandu	9803456789
4	1003	Aayusha Adhikari	Kirtipur	9851987654
5	1004	Niranjan Thapa	Pokhara	9865432109
6	1005	Deepa Magar	Dharan	9741122334
7	1006	Suresh Chaudhary	Biratnagar	9802233445
8	1007	Manisha Ghimire	Chitwan	9813344556
9	1008	Bikram Shah	Butwal	9844455667
10	1009	Kritika Pandey	Hetauda	9867788990

```

INSERT INTO issue (ISBN, mid, issuedate, returndate) VALUES
(100, 1000, '2026-06-01', DEFAULT),
(101, 1001, '2026-06-02', DEFAULT),
(102, 1002, '2026-06-03', DEFAULT),

```

```

(103, 1003, '2026-06-04', DEFAULT),
(104, 1004, '2026-06-03', DEFAULT),
(105, 1005, '2026-06-15', DEFAULT),
(106, 1006, DEFAULT, DEFAULT),
(107, 1007, DEFAULT, DEFAULT),
(108, 1008, DEFAULT, DEFAULT), (109, 1009, DEFAULT, DEFAULT);
SELECT *
FROM issue;

```

Output:

	lid	ISBN	mid	issuedate	returndate
1	5000	100	1000	2026-06-01	NULL
2	5001	101	1001	2026-06-02	NULL
3	5002	102	1002	2026-06-03	NULL
4	5003	103	1003	2026-06-04	NULL
5	5004	104	1004	2026-06-03	NULL
6	5005	105	1005	2026-06-15	NULL
7	5006	106	1006	2026-06-30	NULL
8	5007	107	1007	2026-06-30	NULL
9	5008	108	1008	2026-06-30	NULL
10	5009	109	1009	2026-06-30	NULL

5. Create a view to display all details of the books table

Query:

```

CREATE VIEW book_view AS
SELECT btitle, category, price, nop
FROM book;
SELECT * FROM book_view;

```

Output:

	btitle	category	price	nop
1	Introduction to Algorithms	Computer Science	850	600
2	Learning Web Development	Web Design	320	250
3	Cloud Computing Bible	Infrastructure	750	480
4	Network Security Essentials	Security	580	310
5	Discrete Mathematics	Mathematics	420	290
6	The C++ Programming Language	Programming	690	540
7	Deep Learning Textbook	AI & Data Science	980	620
8	Compiler Design Foundations	Computer Science	510	330
9	The Hobbit	Fantasy Fiction	350	310
10	The Great Gatsby	Classic Literature	280	180

6. Create a view to display book title, category and price only.

Query:

```
CREATE VIEW book_view2 AS  
SELECT btitle, category, price  
FROM book;
```

```
SELECT * FROM book_view2;
```

Output:

	btitle	category	price
1	Introduction to Algorithms	Computer Science	850
2	Learning Web Development	Web Design	320
3	Cloud Computing Bible	Infrastructure	750
4	Network Security Essentials	Security	580
5	Discrete Mathematics	Mathematics	420
6	The C++ Programming Language	Programming	690
7	Deep Learning Textbook	AI & Data Science	980
8	Compiler Design Foundations	Computer Science	510
9	The Hobbit	Fantasy Fiction	350
10	The Great Gatsby	Classic Literature	280

7. Create a view to display books with price greater than 500.

Query:

```
CREATE VIEW price_view AS  
SELECT btitle,price  
FROM book  
WHERE price>500
```

```
SELECT * FROM price_view;
```

Output:

	btitle	price
1	Introduction to Algorithms	850
2	Cloud Computing Bible	750
3	Network Security Essentials	580
4	The C++ Programming Language	690
5	Deep Learning Textbook	980
6	Compiler Design Foundations	510

8. Create a view to display members who have borrowed books.

Query:

```

CREATE VIEW members_borrow AS
SELECT m.mid,m.mname,m.address,m.phone,b.ISBN,b.btitle,i.issuedate,i.returndate
FROM members m
INNER JOIN
issue i
ON m.mid = i.mid
INNER JOIN
book b
ON i.ISBN = b.ISBN
SELECT* FROM members_borrow;

```

Output:

	mid	mname	address	phone	ISBN	btitle	issuedate	returndate
1	1000	Rajesh Khadka	Lalitpur	9812345678	100	Introduction to Algorithms	2026-06-01	NULL
2	1001	Pooja Shrestha	Bhaktapur	9841234567	101	Learning Web Development	2026-06-02	NULL
3	1002	Sandeep Tamang	Kathmandu	9803456789	102	Cloud Computing Bible	2026-06-03	NULL
4	1003	Aayusha Adhikari	Kirtipur	9851987654	103	Network Security Essentials	2026-06-04	NULL
5	1004	Niranjan Thapa	Pokhara	9865432109	104	Discrete Mathematics	2026-06-03	NULL
6	1005	Deepa Magar	Dharan	9741122334	105	The C++ Programming Language	2026-06-15	NULL
7	1006	Suresh Chaudhary	Biratnagar	9802233445	106	Deep Learning Textbook	2026-06-30	NULL
8	1007	Manisha Ghimire	Chitwan	9813344556	107	Compiler Design Foundations	2026-06-30	NULL
9	1008	Bikram Shah	Butwal	9844455667	108	The Hobbit	2026-06-30	NULL
10	1009	Kritika Pandey	Hetauda	9867788990	109	The Great Gatsby	2026-06-30	NULL

9. Create a view to display the most expensive book.

Query:

```

CREATE VIEW expensive_view AS
SELECT *
FROM book
WHERE price = (SELECT MAX(price) FROM book)
SELECT * FROM expensive_view ;

```

Output:

	ISBN	btitle	category	price	nop
1	106	Deep Learning Textbook	AI & Data Science	980	620

10. Create a view to display books that have never been issued.

Query:

```

CREATE VIEW never_issued AS
SELECT * FROM book

```

WHERE ISBN NOT IN(SELECT ISBN FROM issue)

SELECT * FROM never_issued;

Output:

Results		Messages			
	ISBN	btitle	category	price	nop
1	110	Chemistry	Science	650	500

11. Increase price of all books by 5%.

Query:

UPDATE book

SET price = price + (price*0.05)

SELECT * FROM book

Output:

Results		Messages			
	ISBN	btitle	category	price	nop
1	100	Introduction to Algorithms	Computer Science	892	600
2	101	Learning Web Development	Web Design	336	250
3	102	Cloud Computing Bible	Infrastructure	787	480
4	103	Network Security Essentials	Security	609	310
5	104	Discrete Mathematics	Mathematics	441	290
6	105	The C++ Programming Language	Programming	724	540
7	106	Deep Learning Textbook	AI & Data Science	1029	620
8	107	Compiler Design Foundations	Computer Science	535	330
9	108	The Hobbit	Fantasy Fiction	367	310
10	109	The Great Gatsby	Classic Literature	294	180
11	110	Chemistry	Science	682	500

12. Create a view to display the total number of books borrowed by each member.

Query:

CREATE VIEW member_book_count AS

SELECT m.mid, m.mname, COUNT(i.ISBN) AS total_books_borrowed

FROM members m LEFT JOIN

issue i

ON m.mid = i.mid

GROUP BY m.mid, m.mname;

SELECT *

FROM member_book_count ;

Output:

Results		Messages	
	mid	mname	total_books_borrowed
1	1000	Rajesh Khadka	1
2	1001	Pooja Shrestha	1
3	1002	Sandeep Tamang	1
4	1003	Aayusha Adhikari	1
5	1004	Niranjana Thapa	1
6	1005	Deepa Magar	1
7	1006	Suresh Chaudhary	1
8	1007	Manisha Ghimire	1
9	1008	Bikram Shah	1
10	1009	Kritika Pandey	1